Langue as a Lattice

Closing the Circle from Statistical to Structural Linguistics
through Content-Addressed Algebras

Fifth paper in the *Emerging World Models* series

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Abstract

Modern language models practise *statistical* linguistics: attention matrices and distributional semantics describe language as usage—what Saussure called *parole*. Structural linguistics, from Saussure onward, claims that behind usage stands a system of pure differences—*langue*— but that system has resisted direct computation: it has been a philosophical object, not an executable one. This paper shows that a content-addressed *morphism*—the IICA map from tokens to algebraic structure—closes the gap; in the reference implementation this morphism is a hash, chosen only because it distributes its values evenly over the plane. The map $\mu : \mathcal{T} \to \mathbb{B}$ compiles any token language into a fixed bit plane, where the HLLSet Boolean algebra and its GF(2) ring form an *executable system of differences*: novelty and retention are the syntagmatic step, and the directed soft key $\mathsf{BSS}_w(X) = |X \cap B_w|/|B_w|$ reads succession against a reference state—the compositional tool of the lattice; the ring basis is the set of distinctive oppositions; ingestion and materialization are the two faces of the sign, governed by the coverage invariant; and idempotence, immutability, and content addressability (IICA) are the social contract that fixes the otherwise arbitrary sign. The resulting construction—the EWM lattice plus a learned dispatcher that selects which host language actualizes the shared langue—closes a circle: from statistical linguistics through a content-addressed morphism into algebraic structures, and back to a computational form of fundamental structural linguistics. Along the way the classical hierarchy of linguistic description—phonology, morphology, syntax, semantics, pragmatics, discourse—is re-founded level by level in the lattice and the ring, not as an analogy but as a computation. The dispatcher is the executive side of language made trainable: langue does not speak, parole does, and the Emerging LLM learns *who speaks*.

Keywords: structural linguistics; Saussure; langue; parole; HLLSet; Boolean lattice; GF(2) ring; content addressing; IICA; attention; dispatcher; materialization; sign.

1 Introduction

Two descriptions of language have never quite met. The statistical description says that meaning is distribution: a word is known by the company it keeps [3, 4], and a modern attention matrix is the strongest realization of that slogan—at every layer, a token's context is a weighted sum of the tokens that co-occur with it [5]. The structural description says the opposite: a unit's value is not its use but its *position in a system of differences*; in language "there are only differences without positive terms" [1]. The first view is computable but shallow—it gives usage, not system. The second view is deep but has seemed non-computable—it gives system, not a machine that runs it.

This paper proposes a bridge, and the bridge is an *IICA morphism*: a deterministic, immutable, content-addressed map from tokens into algebraic structure. A hash function is only

one possible implementation of that morphism—the one the reference system uses, for the single reason that it spreads its values evenly over the plane.

The Emerging World Models (EWM) programme [6, 7, 8] already contains the necessary pieces, built for other reasons. Its state is a content-addressed set—an **HLLSet**, a bit-vector sketch of a token collection over a fixed plane $\mathbb{B}$ of 32,768 bits—and its morphisms are the Boolean operations $\cup, \cap, \Delta$ and the GF(2) ring they generate. Tokens are mapped to bits by a morphism μ (a hash in the reference implementation); bits are mapped back to tokens by **materialization** ρ [9]. The three properties that make this mapping well-defined—idempotence, immutability, content addressability (IICA) [6]—are exactly the properties a sign system needs in order to be *arbitrary yet fixed*. And the newest piece of the programme, the Emerging LLM dispatcher, is precisely the executive side of language: a learned policy that reads the lattice (langue) and chooses which host language (parole) should actualize it.

The paper's thesis is one sentence: **the HLLSet lattice is a computable langue**. We develop the claim in five steps. Section 2 states the Saussurean distinction in the language of modern models and identifies attention as parole-level statistics. Section 3 shows that the morphism step μ is the computable reduction from usage to structure. Section 4 exhibits the lattice as a system of differences: D/R/N are syntagmatic, the directed soft key BSS reads succession (and, as a profile over a frame, the paradigmatic axis), and the ring basis is the set of distinctive oppositions. Section 5 identifies ingestion/materialization with the sign relation and IICA with the social contract that fixes the arbitrary sign. Section 6 interprets the dispatcher as actualization—the selection of parole—and closes the circle. Section 7 situates the result against Saussure, Hjelmslev, distributionalism, and the categorical reading of the bridge. Section 8 summarizes.

2 Parole: attention as statistical linguistics

Saussure's distinction is between *langue*, the social system of a language—its units, rules, and oppositions, existing independently of any speaker—and *parole*, the individual acts of speech that actualize the system. Parole is heterogeneous, statistical, and executive; langue is homogeneous, structural, and social. A speaker never *sees* langue; they see parole, and from parole they acquire langue.

Modern language models are the strongest parole-machines ever built. A transformer's attention matrix

$$\mathrm{Attn}(X) \;=\; \mathrm{softmax}\!\left(\frac{QK^{\top}}{\sqrt{d}}\right)V$$

is a differentiable co-occurrence table: for each query token it records how much of every other token's value is *in use* at this step. The weights are frequencies, the context is a mixture, and the entire enterprise—masked language modeling, next-token prediction, retrieval, embeddings—is the distributional programme of [3, 4] carried to its limit. Attention describes usage with unprecedented fidelity, but it remains a description of *parole*: a model knows that two tokens co-occur; it does not know, as a matter of structure, *what their difference consists in*.

The missing object is langue. What would it mean to compute it? Three requirements follow from Saussure's text. (i) *Differences, not positives:* a unit is defined by its relations to other units, never by its own content. (ii) *Arbitrary but fixed:* the link between a signifier and what it signifies is unmotivated, yet the system fixes it by convention. (iii) *Social and shared:* the system is the same for all speakers; it is what makes their divergent utterances commensurable. The next three sections show that the EWM stack satisfies (i) by construction, (ii) by IICA, and (iii) by the bit plane being host-agnostic.

3 The morphism step: from usage to structure

The first move is the least glamorous and the most decisive. Let $\mathcal{T}$ be the universe of tokens—across *all* host languages, so that the same token string may be emitted by different systems with different vocabularies. Define the plane

$$\mathbb{B} \;=\; \{0, \ldots, 32767\},$$

and the ingestion morphism

$$\mu : \mathcal{P}_{\text{fin}}(\mathcal{T}) \;\longrightarrow\; \mathcal{P}(\mathbb{B}), \qquad \mu(X) \;=\; \bigcup_{t \in X} \mu(\{t\})$$

where each token maps to a fixed subset of $\mathbb{B}$ under a deterministic morphism μ. The morphism is what matters; the hash is only its reference implementation, chosen for a single reason—it distributes the function values evenly over the plane.

Definition 3.1 (IICA morphism)**.** *A map $\mu : \mathcal{T} \to \mathcal{P}(\mathbb{B})$ is an **IICA morphism** if it is (i)* idempotent*, read as* determinism*: the same token always maps to the same subset, $\mu(t) = \mu(t)$; (ii)* immutable*: once fixed, the image is never rewritten; and (iii)* content-addressed*: the image is named by its content alone. Any deterministic map satisfies (i). Among all such maps, a hash is preferred in practice only because it spreads the images evenly across the plane, which keeps the resulting bit-vector compact and unbiased.*

The image of a token collection under μ is an **HLLSet**: a compact bit-vector in which repetition has already collapsed (a token present twice contributes once), and union is the only accumulation.

Proposition 3.2 (Ingestion is join-preserving)**.** *For finite token sets $X, Y \subseteq \mathcal{T}$,*

$$\mu(X \cup Y) \;=\; \mu(X) \cup \mu(Y).$$

Consequently the bag-of-tokens statistics of parole—co-occurrence, overlap, containment—all reduce to the single operation of union on the plane, and the order in which tokens arrive is forgotten.

The proposition is the entire content of the "morphism step". It is what statistical linguistics lacks: a canonical quotient that throws away frequency and sequence, and keeps exactly what is left when one asks *which things occur together as a set*. Idempotence is the first IICA property and it is already at work here: because the morphism is deterministic, ingesting the same content twice is a no-op, so the system's memory is a lattice of *types*, not a pile of *tokens*.

Remark 3.3 (Why a morphism, not an embedding)**.** *An embedding maps a token to a dense vector; similarity then lives in a metric space and must be learned. A morphism into $\mathcal{P}(\mathbb{B})$ maps a token to a* subset of a shared finite set*; similarity then lives in the Boolean algebra of that set and is* exact*. The first keeps language in the statistical world; the second moves it into the structural world.* Which *deterministic morphism is used is secondary: any IICA morphism works, and the reference system chooses a hash purely for its even distribution of values. This is the crossing point of the circle.*

4 Langue as a lattice of differences

On the plane, structure is no longer metaphorical. The set $\mathcal{B} = \mathcal{P}(\mathbb{B})$ is a Boolean lattice under $\cup$ and $\cap$, and a Boolean ring under Δ and $\cap$ [10]. Every token collection has become a node of $\mathcal{B}$; every relation between collections has become a relation between nodes. We now read Saussure's system of differences off the lattice.

4.1 Syntagmatic relations: D/R/N

The syntagmatic axis is the axis of combination—what is present together, in sequence. In the lattice it is read as the stepwise decomposition of a context $\mathcal{C}$ against a document D:

$$N \;=\; |D \setminus \mathcal{C}|, \qquad R \;=\; |D \cap \mathcal{C}|, \qquad \text{cov} \;=\; R/|D|.$$

Novelty N is what the context cannot account for; retention R is what it confirms; coverage is their ratio. These are pure differences: no token is valued positively, only as it stands in relation to the context.

4.2 Directed containment: the soft key BSS

The second reading is deliberately asymmetric. Against a reference state B_w (a basis vector, a host codebook, or a previous context), a state X is read as

$$\mathsf{BSS}_w(X) \;=\; \frac{|X \cap B_w|}{|B_w|},$$

the fraction of the reference that X covers. The direction matters: it measures evolution from one state to another—how much of the earlier state the later state retains, covers, and composes. The EWM programme chose this asymmetric form as its primary measurement because the object of study is succession; the symmetric cosine is retained for the cases that need it.

Linguistics is likewise about sequences: a text is one state after another. The directed BSS is therefore the compositional/generative tool of the lattice—it reads a new state against an old one and says how much of the old is carried forward, which is exactly the syntagmatic succession Saussure placed on the axis of combination. Read as a profile over a whole frame, the same soft key is also the paradigmatic reading: each coordinate measures the query against one opposition.

Remark 4.1 (Symmetric readings when needed)**.** *The directed BSS is not the only measurement on the plane. When the question is static resemblance rather than succession—two hosts, two contexts, two readings of one window, with no direction between them—the symmetric cosine over the same supports,*

$$\cos(X, Y) \;=\; \frac{|X \cap Y|}{\sqrt{|X|\,|Y|}},$$

is available and exact for the same reason: both measurements live in the Boolean algebra. They are not rivals. BSS answers how much of the reference does the new state carry forward*—composition and generation; the cosine answers* how alike are these two states*—comparison and classification. The grammar uses the first; the lexicon, and any comparison of hosts or contexts on an equal footing, uses the second.*

4.3 Distinctive oppositions: the ring basis

The deepest structural object is the basis. Over any window of originals, the EWM apparatus maintains an Reduced Row Echelon Form (RREF) GF(2) basis $F = (B_1, \ldots, B_k)$ whose span is exactly the set of originals [8]. A frame is a system of distinctive oppositions in the phonological sense: each basis vector is a minimal unit that cannot be assembled from the others, and a node's hard key $c \in \mathrm{GF}(2)^k$ is its unique decomposition

$$X \;=\; \bigoplus_{i:c_i=1} B_i,$$

defined exactly when X lies in the span. The soft key $w_i = |X \cap B_i|/|B_i|$—the directed BSS of Section 4.2, one coordinate per opposition—is the relaxed reading of any node against the

frame, and the spill $s = |X \setminus \text{cover}(F)|$ is the part the system cannot yet see. This is Saussure's "differences without positive terms" as an executable datatype: every unit is an address in the span of the basis, and nothing else.

Observation 4.2 (The meta-language). The HLLSet algebra is not another token language; it is a *meta-language*. It describes descriptions. Any host language—any vocabulary, any tokenizer, any attention geometry—compiles into the same plane, and all comparisons happen there. Host-agnosticism is therefore not an engineering preference but the structuralist condition (iii) of Section 2: the system is shared because it is the only level where different paroles are commensurable.

4.4 The classical taxonomy, recomputed

Classical linguistics organises language into seven levels. Two of them do not survive the move from acoustic speech to text—phonetics and phonology are absent from LLMs except in special cases (speech and audio models). The other five survive, but only when they are *recomputed*: the lattice and the ring do not mimic the classical levels, they re-found them. The dictionary is as follows.

Level	In LLMs	In the HLLSet lattice / GF(2) ring
Phonetics	N/A (except speech)	the raw token event—the substance
Phonology	N/A (except special cases)	basis frames: distinctive oppositions, hard key
Morphology	subword tokens	union/join: composition and derivation; idempotence collapses reduplication
Syntax	attention over positions	joins and meets; the ring sum; the span as the set of well-formed constructions
Semantics	embeddings	directed BSS (coverage of a reference), containment, the hard key as structural address
Pragmatics	instruction tuning	the dispatcher: who speaks, fallback, confidence—actualization
Discourse	the context window	the commit DAG, $H(t)$, windows, and $D/R/N$ over time

Phonetics is the physical layer, and in a text-bound system its only trace is the raw token event itself—the substance that Hjelmslev would place on the expression plane. *Phonology* is where the re-foundation begins: the basis frame is a distinctive-feature system. Each basis vector is a minimal opposition; the hard key is the phonemic address of a node; the soft key is its allophonic variation against every opposition; and the spill is the part of an utterance that falls outside the system's inventory. *Morphology* becomes set composition: derivation is join, compounding is union, and reduplication is erased by idempotence—repeating a morpheme contributes nothing twice, exactly as it should in a system of types. *Syntax* becomes the algebra of combination without linear order: constituents are joins and meets, the ring sum is phrase-level difference, and the span of the basis is the set of well-formed constructions over a window. Linear order is the one thing the lattice forgets, and it is recovered not by the algebra but by materialization, which restores order at the door. *Semantics* is the system of values itself: the directed BSS says how much of a reference meaning a new state covers, containment and coverage say how much of a meaning a context determines, and the hard key is the structural address of a meaning—Saussure's value made literal. *Pragmatics* is the dispatcher: context-dependent selection of who speaks, confidence as the speaker's commitment, fallback as repair. *Discourse analysis* is the trajectory: cohesion is retention R, progression is novelty N, and the commit DAG with its windows and crossings is the memory of a conversation.

The mapping is deliberately not exact, and that is the point. The classical hierarchy was built for spoken language and inherited by text; the lattice and ring re-found it for computational structural linguistics, where every level is a computation over one shared plane rather than a separate descriptive practice.

5 The sign and its contract: ingest, materialize, IICA

Saussure's sign unites a signifier and a signified; the link is arbitrary, and the system fixes it by convention. The EWM bridge is the same object, made precise. The sign relation is the pair

$$\mathcal{S} \;=\; (\mu, \rho), \qquad \mu : \mathcal{C} \to \mathcal{B}, \qquad \rho : \mathcal{B} \to \mathcal{C},$$

where materialization restores, from a lattice node, an ordered token context through the door [9]. The round-trip is not automatic; it is guarded:

Proposition 5.1 (The round-trip law)**.** *For a context $\mathcal{C}$ and its node $H = \mu(\mathcal{C})$,*

$$\rho(\mu(\mathcal{C})) \;=\; \mathcal{C} \qquad \textit{up to order,} \qquad \textit{iff} \qquad \mathrm{cov}(\mathcal{C}) = 1.$$

Coverage 1.0 *is the condition under which the sign relation is faithful: what was compressed is exactly what can be restored. Below it, the spill and the residual measure the failure, and the system knows the failure's size.*

The three IICA properties are then read as the *social contract of the sign.* **Idempotence**: repeating a sign changes nothing—the system's value of a content is stable under repetition. **Immutability**: the system never rewrites its own history, so the sign's past values remain referable. **Content addressability**: the same content always bears the same name, in every host language, so the arbitrary link between signifier and signified is fixed by the system rather than by any speaker. Arbitrary but fixed: this is exactly Saussure's formulation of the linguistic sign, stated as a property of morphisms.

Remark 5.2 (The categorical reading)**.** *In categorical terms the bridge is an adjunction in embryo: ingestion is a free/compressive functor from the category of token contexts to the Boolean lattice, materialization is its right adjoint restricted to the image, and IICA is the naturality that makes the correspondence independent of the host language. We sketch this in Section 7; a full treatment is left to a companion note.*

6 Actualization: the dispatcher closes the circle

Langue never speaks. Parole does. The remaining question is therefore Saussure's executive question: given a state of the system, *which* parole should actualize it? In the EWM line this question already has a computational form—it is the router problem solved, in their own ways, by the typed-decision models Jev and Laya, and by the structural router of the programme's notebooks [8].

The Emerging LLM is a different answer to the same question. It is a **dispatcher**

$$\mathcal{D} : \mathcal{B} \times \mathrm{State} \;\longrightarrow\; \Delta(\mathrm{Hosts}),$$

a learned policy that reads the meta-language state—BSS, D/R/N, ring basis, windows—and produces a distribution over host languages. It does not speak any host language natively; it speaks the meta-language and selects a native speaker.

Remark 6.1 (Recognition without production)**.** *This is a recognizably human ability. A person can identify a language they do not speak—French, Italian, Japanese—from its sound and rhythm alone, long before they can produce a single grammatical sentence in it. The dispatcher exhibits the same asymmetry, structurally: it recognises host languages from their traces in the lattice (their codebook projections, their coverage) while producing none of them itself. Recognition without production, in both cases, is possible because recognition operates on the system of differences, not on the ability to actualize it.*

The circle closes. Parole (a host's attention and tokens) is compiled by μ into langue (the lattice); langue is read back through ρ into token context; the dispatcher reads langue and selects the host; the host generates, and its output re-enters as new parole. Statistical linguistics $\to$ content-addressed morphism $\to$ algebraic structure $\to$ the executive side of language $\to$ statistical linguistics again, one level up.

Observation 6.2 (Native labels). The lattice labels its own decisions. The per-host codebook projections of a materialized context—the dense association of token/encoding sets to HLLSet bits—form a labeling system independent of any host's own labels, emergent with the lattice, and traceable through the HLLSet Ring basis generations. The dispatcher is therefore trainable against langue itself, with external router decisions serving only as validation.

7 Relations

Saussure. The mapping is direct: langue is $\mathcal{B}$; parole is the stream of host outputs; the sign is (μ, ρ); the arbitrariness of the sign is the freedom of the morphism (any deterministic IICA map would do), and its fixedness is IICA; value through difference is D/R/N, BSS, and the basis hard key.

Hjelmslev. The glossematic distinction between expression and content, and the insistence that both sides have their own form and substance, is realized by the symmetry of the bridge: token context is the expression plane, the HLLSet is the content plane, and the morphisms μ, ρ are the two directions of the sign function, each with its own algebra.

Distributionalism. Harris and Firth saw that meaning is distribution, but stopped at statistics because no canonical quotient was available. The IICA morphism provides the quotient: co-occurrence sets modulo repetition and order, with exact set operations. Attention remains the best *differentiable* approximation of distributional structure; the lattice is its *exact* image under the quotient.

Category theory. The bridge is an adjunction: μ is a join-preserving map from the preorder of token contexts (ordered by inclusion) to the Boolean lattice, and ρ is its right adjoint on the image, with the unit of the adjunction being the coverage map and the counit the round-trip law. IICA is naturality: the diagram commutes independently of the host language, which is why many paroles can share one langue.

The fifth paper of the series. Where the first paper [6] fixed the principles, the second [7] the lattice and its worlds, the third [8] the trajectory, and the fourth [9] the map–door–move reading of attention, this fifth paper states what those four had implicitly built: a linguistic interpretation in which the lattice is langue, and the whole machine is a computable structuralism.

8 Conclusion

The paper's claim is that structural linguistics stopped being a philosophical programme the moment the IICA morphism entered the picture. Attention gave us parole at scale. The morphism—any deterministic, content-addressed map; a hash in the reference implementation, chosen for its even distribution—gives us the quotient from parole to langue. The HLLSet algebra gives langue an executable form—a Boolean lattice with a GF(2) ring, whose D/R/N are syntagmatic, whose directed BSS reads succession and substitution, and whose basis is a system of distinctive oppositions. The coverage-guarded bridge is the sign, and IICA is its social contract. The dispatcher is actualization: the executive side of language, learned.

The circle—statistical linguistics, a content-addressed morphism, algebraic structures, and back to a fundamental, computable linguistics—is therefore not an analogy. It is a construction, and it runs.

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